ClaimsGuru | Enterprise Health Reimbursement Platform

Complete Product, Architecture & Onboarding Guide

1. Executive Overview: What is ClaimsGuru?

ClaimsGuru is an enterprise-grade, AI-powered health insurance claim intake, audit, and adjudication platform engineered by WaferWire Cloud Technologies (WCT). Operating natively on Microsoft Azure, ClaimsGuru bridges patients, healthcare providers (hospitals), and insurance Third-Party Administrators (TPAs) into a unified, high-speed ecosystem.

Historically, health insurance reimbursement in India has required policyholders to mail stacks of physical paper bills, wait weeks for manual keypunching, and endure high technical rejection rates. ClaimsGuru ingests messy, scanned, multi-page hospital invoices, handwritten discharge summaries, and pharmacy receipts, and autonomously transforms them into structured, verified, and audited insurance dossiers in under 3 minutes.

2. The Problem in the Indian Health Insurance Ecosystem

Industry Pain Point	Real-World Impact on Patients & Insurers	ClaimsGuru Autonomous Fix
15 to 30 Day Delays	Physical bill courier, manual data entry, and postal back-and-forth stall reimbursements for weeks.	Instant digital intake & OCR reduces processing cycle to under 3 minutes.
>18% Technical Rejections	Claims rejected due to patient typos, missing invoices, or mixing family member records.	Fail-Fast Identity Consensus Gate validates & rejects mismatched papers pre-submission.
Surprise Tariff Cuts	Hospitals bundle hundreds of non-payable consumables; patients face unexpected cuts.	Automated IRDAI Schedule I–IV engine itemizes approved vs. deducted amounts upfront.
Policy Clause Complexity	Complex rules: 1% Room Rent caps, co-payments, and Pre-Existing Disease (PED) waiting periods.	Autonomous Policy Schedule Engine cross-audits bills directly against plan limits.

3. Multilingual Accessibility: 14 Indian Languages & Bi-directional RTL

To empower policyholders across all Indian states and vernacular regions, ClaimsGuru provides complete native localization across 14 constitutionally recognized languages without reliance on real-time external machine translation:

Multilingual Capability	Enterprise Technical Implementation & Regional Coverage
14 Native Languages	Complete UI coverage: English, Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia, Assamese, Urdu, and Sanskrit rendered in native regional scripts.
Urdu RTL Support	Automatic bi-directional text flow and layout switching (dir='rtl') for seamless Urdu reading and form completion.
Persistent Preference	User language selection is saved across sessions in localStorage (claimsguru_lang) and synchronized to the HTML root element.
WCAG AA Accessibility	Guaranteed high-contrast compliance (contrast ratio ≥ 4.5:1) across all script systems with 44×44px touch targets on mobile and desktop.

4. How ClaimsGuru Works: The 10-Stage Autonomous Pipeline

Every incoming insurance submission is processed through an automated, asynchronous ten-stage clinical pipeline:

Pipeline Stage	Technical Workflow & Autonomous Capabilities
1. Multi-Format Intake	Ingests single/multi-page PDFs, JPEG, PNG, and camera photos up to 25 MB. Computes a set-based SHA-256 fingerprint to immediately block duplicate submissions.
2. Identity Gate	Cross-checks patient name, age, and gender across all documents. Automatically isolates and rejects mismatched family member or third-party files before processing.
3. Vision & Table OCR	Combines Azure AI Document Intelligence with custom vision heuristics to parse complex multi-page hospital invoices with hundreds of line items and taxes.
4. Clinical NER & Codes	Extracts diagnoses and surgical procedures via Biomedical LLMs and FAISS vector indices, automatically mapping them to official ICD-10-CM and CPT codes.
5. Policy Plan Engine	Extracts Sum Insured (SI), Room Rent Limits, Co-Payment Clauses, PED waiting periods, and specific surgical sub-limits (maternity, cataract, AYUSH).
6. Policy Cross-Audit	Compares hospital billing categories against policy limits. Calculates proportional room rent deductions and separates non-payable consumables under IRDAI guidelines.
7. 11 IRDAI Rules & Risk	Evaluates the claim against 11 mandatory IRDAI validation rules and executes XGBoost/LightGBM risk models to predict approval odds and flag rejection triggers.
8. AcroForms & Dossier	Auto-populates fillable IRDAI Claim Form (Part A & Part B) PDF AcroForms with digital touchscreen signature pads and downloadable clinical audit dossiers.
9. Missing Policy Flow	Allows patients to upload a missing policy schedule later ('Update Policy Doc'), seamlessly recalculating coverage limits without needing to re-upload hospital bills.
10. TPA Review Portal	Provides hospital and TPA adjusters with review consoles (/[orgSlug]/review) to approve, reject, or request specific additional records (DOCUMENTS_REQUESTED).

5. Security, Privacy & Zero Egress Architecture

Security Pillar	Technical Control & Architecture Implementation	Regulatory Standard
Zero Public AI Egress	Patient Health Information (PHI) and PII are processed strictly within isolated Azure tenant boundaries. No customer data is ever shared with or used to train commercial public AI models.	HIPAA & Enterprise BAA
DPDP Act, 2023 Compliance	Explicit digital consent verification, data minimization, automated retention schedules, and one-click right to personal data erasure.	DPDP Act, India 2023
Bank-Grade Cryptography	End-to-end TLS 1.3 in transit; AES-256 bit Transparent Data Encryption (TDE) at rest across all databases and Azure Blob storage containers.	SOC 2 Type II / ISO 27001
Immutable Audit Logging	All claim events, identity gate evaluations, reviewer overrides, and document modifications are permanently recorded in dbo.audit_logs with UTC timestamps.	IRDAI Audit Compliance

6. Complete Technical Stack Specification

ClaimsGuru is architected as a modern, high-throughput microservices ecosystem on Microsoft Azure:

Architecture Layer	Technology & Version	Role & Architectural Responsibility
Frontend Portal	Next.js 14, React 18/19, Tailwind CSS, Lucide Icons	WCAG AA accessible self-service portal, TPA adjudication consoles, responsive from 375px mobile to 1366x768 desktop.
API Gateway	Python 3.11, FastAPI (v0.128.8), Uvicorn uvloop	High-throughput asynchronous REST gateway handling claim lifecycle, document uploads, and real-time state polling.
Async Workers	Celery (v5.4.0), Azure Cache for Redis, Azure Service Bus	Isolated queues (ocr_queue, parser_queue, default_queue) ensuring compute-heavy OCR never blocks API traffic.
Relational DB	Azure SQL Database (MS SQL Server), SQLAlchemy 2.0, Alembic	ACID-compliant relational persistence for claims, policy records, line items, and audit trails.
Generative AI / LLM	Azure OpenAI Service (GPT-4o, API 2024-11-20)	Private, zero-egress LLM engine for clinical NER, medical term normalization, and policy contract analysis.
Computer Vision OCR	Azure AI Document Intelligence (Form Recognizer v3.3.0)	Deep document layout analysis, bounding box extraction, and multi-page table detection for Indian hospital bills.
Identity & Security	Microsoft Entra External ID (CIAM), Azure Key Vault	OAuth2 / OIDC Single Sign-On with RS256 JWT validation, hardware-backed HSM encryption, and DPDP/HIPAA compliance.

7. Essential Health Insurance Domain Glossary

Insurance Term	Definition & How It Applies in ClaimsGuru
TPA (Third-Party Administrator)	Licensed entity managing claim adjudication between hospitals, insurers, and patients (e.g. Medi Assist, Vipul, Paramount, Vidal).
IRDAI	Insurance Regulatory & Development Authority of India. Sets statutory claim rules, Part A/B forms, and non-payable expense limits.
Sum Insured (SI)	The annual financial ceiling up to which the insurance company covers the patient's medical expenses.
Room Rent Capping	Clause capping daily room charges (e.g. 1% of SI). Upgrading to a costlier room triggers proportional deductions on doctor & OT fees.
Co-Payment	Mandatory cost-sharing (e.g. 10% or 20%) that the policyholder must pay from their own pocket before the insurer covers the rest.
PED (Pre-Existing Disease)	A medical ailment existing prior to policy issuance. Subject to a 24- to 48-month waiting period before claims become eligible.
AcroForm	Interactive form architecture embedded in PDFs that ClaimsGuru auto-populates for digital IRDAI Part A & B signing.
ICD-10-CM & CPT	International diagnostic (ICD-10) and surgical/procedural (CPT) medical coding standards mapped automatically by ClaimsGuru AI.

Welcome to ClaimsGuru! This platform represents a generational leap in health insurance technology. For technical specifications and API contracts, refer to docs/architecture.md and the interactive Swagger docs at http://localhost:8000/docs. For questions or architecture reviews, connect with the WaferWire Cloud Technologies engineering leads.